MILESTONE 1 / NETWORKING KNOWLEDGE AND SKILL DEVELOPMENT

AI-Powered Network Diagnostics and IT Risk Assessment Agent

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Project objective

When an order-entry health check fails, the agent investigates whether the local DNS fixture can return the expected address, whether a TCP connection can be made to the service, and how the HTTP endpoint responds. Codex reads the initial **diagnose-order-entry** Skill and runs the probes. Python provides the two local services used to control the experiment.

How networking knowledge became instructions

Course concept	Decision encoded in the Skill
DNS and A records Lecture 5	Send an A query for orders.test. directly to the fixture. Node's built-in DNS resolver supports the lab's custom UDP port. Check the returned IPv4 address and distinguish a negative DNS response from a timeout before investigating the service.
Loopback, sockets and TCP ports Lectures 4, 8 and 10	Use curl to attempt a TCP connection to the fixed lab endpoint. Use ss, filtered to port 8765, to verify whether the expected IPv4 socket is listening. Correlate listener state with any connection failure.
HTTP response semantics Lecture 6	Read curl's process exit code and the HTTP status separately. HTTP 200 shows the health endpoint replied. An HTTP error also shows that a responder answered. Curl's 000 write-out means no HTTP status was received.

Scope of the probes

Both fixtures bind to 127.0.0.1 in the same Linux environment as the agent. The DNS fixture receives the A query on UDP port 15353. Curl uses `--resolve` to set the mapping for the HTTP request on TCP port 8765. That isolates the service check from the machine's usual DNS resolution path. The results apply to the local health endpoint. A successful response leaves the actual order workflow untested.

Course materials consulted

Lecture 4, Application Layer Part I (system interface and sockets). Lecture 5, Application Layer Part II (DNS). Lecture 6, Application Layer Part III (HTTP). Lecture 8, Transport Layer Part I (TCP and ports). Lecture 10, Network Layer Part I (IP addressing and loopback).

Diagnosing a stopped HTTP listener

Skill loading and actual outputs

Codex followed the packaged SKILL.md workflow in the remote Linux lab. The demonstration used explicit file loading in ChatGPT Work. The ZIP includes the full file-read output, command logs, screenshots of the recorded outputs, and the agent's assessment.

Run	DNS A answer	curl result	ss listener check
A Healthy	127.0.0.1	HTTP 200, exit 0	Listening on 127.0.0.1:8765
B HTTP stopped	127.0.0.1	No HTTP status, exit 7	No matching listening socket
C HTTP restored	127.0.0.1	HTTP 200, exit 0	Listening on 127.0.0.1:8765

Verification and conclusion

Only the HTTP fixture was stopped for run B. The DNS fixture log shows a successful A query in each run. The HTTP logs show successful GET requests before the stop and after the restart. The missing socket in B and the recovery in C support the agent's conclusion that the stopped listener caused the controlled outage. Any impact on order processing remains conditional on the business using this endpoint.

Progress and next steps

Completed. Set up the loopback lab, drafted the initial Skill, and captured the healthy, stopped-listener, and restored runs with raw evidence and an agent interpretation.

Setup issues. The Work environment lacked nslookup and dig, so the lab uses Node's built-in DNS resolver for the direct query. Curl and ss are the two existing networking utilities demonstrated. Default ss socket inspection hit a permission error during setup. The Skill uses the verified proc-based IPv4 check.

Next. Document an NXDOMAIN case and an HTTP 503 case, then verify what the agent concludes in each. Keep a new condition separate for a later evaluation and prepare the workflow for the live demonstration.

Team information and contributions

Nikhil Hemendra Varghese is the sole team member. Nikhil defined the project scope in the proposal, supplied the course and milestone requirements, and directed the milestone work. Codex assisted with the initial Skill, fixture implementation, diagnostic runs, and preparation of the evidence package.